

Aakash KT

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Experience

Dec 2025 - **Senior Graphics Research Engineer**, *Huawei IRC*, Netherlands.

May 2026 Part of the Game Rendering Lab, with projects on identifying and resolving bottlenecks in real-time rendering of mobile games.

Work overview:

- Analyzing GL calls with RenderDoc, inspecting rendering layers and draw calls to determine optimizations.
- Intercepting OpenGL calls to improve the rendering fidelity or performance.
- Using Vulkan API to inject custom rendering passes in Unity 3D engine via plugins.
- GPU-CPU-NPU synchronization for ML graphics workloads like neural transparency.
- Optimizing shaders by using correct memory layout, shared memory and warp intrinsics.

Jun 2024 - **Senior Graphics Research Engineer**, *Qualcomm*, India.

Oct 2025 Part of the graphics research team, working on future GPU architecture research.

Work overview:

- Validate new hardware and driver features like Shader Execution Reordering (SER) and Work Graphs by building functional prototypes of the GPU stack.
- Cycle-accurate simulation of functional prototype to get performance estimates.
- Collaboration across driver, compiler and hardware teams to bring the prototype to production.
- DirectX 12 driver feature bring-up and performance testing suites, by designing tests with increasing complexity and complete feature coverage.
- Exploring acceleration structures (BVH) and their hardware realization for improved ray-tracing efficiency. Patent filed in this space.

Aug 2022 - **Research Scientist Intern**, *Meta Reality Labs*, Pittsburgh, USA.

Jan 2023 **Manager:** Dr. Giljoo Nam

Part of the Codec Avatars team, with a project on realistic and efficient rendering of human hair.

Work overview:

- Investigating light transport theory, specifically for hair rendering.
- Implementing physically based and approximation based hair rendering baselines in C++, OptiX and CUDA.
- Deriving a novel solution based on theory investigations and experimental observations of baseline methods to achieve state-of-the-art rendering capabilities.
- Designing an on-the-fly training and inference strategy and setting an appropriate target signal for a small neural network (deep learning) to work in combination with physically based rendering.
- Writing up a clear document for publication (work published in EGSR 2023).

Aug 2017 - **Research Assistant**, *CVIT, IIIT Hyderabad*, India.

Aug 2020 **Advisor:** Dr. P. J. Narayanan

Neural Rendering for specific usecases and appearance editing from captured photographs. Additional projects on depth estimation from focus images and explored deep generative models for domain adaptation.

Education

- Aug 2020 - **Ph.D in Computer Science, IIIT Hyderabad.**
May 2024 **Advisor:** Prof. Dr. P. J. Narayanan, **CGPA:** 8.0/10.0
Thesis on accelerating physically based rendering via analytic solutions and efficient neural approximations of the rendering equation.
- Aug 2015 - **BTech and MS by Research in Computer Science, IIIT Hyderabad.**
Aug 2020 **Advisor:** Prof. Dr. P. J. Narayanan, **CGPA:** 7.21/10.0
Relevant courses: Computer Graphics, Computer Vision, Digital Image Processing, Artificial Intelligence, Statistical Methods in AI, Optimization Methods, Advanced Computer Networks, Operating Systems, Software Engineering.

Selected Publications

- Eurographics 2025 **Linearly Transformed Spherical Distributions for Interactive Single Scattering with Area Lights**, *CGF, Full Paper*.
Aakash KT, Ishaan Shah, P. J. Narayanan
- SIGGRAPH Asia 2023 **Combining Resampled Importance & Projected Solid Angle Samplings for Many Area Light Rendering**, *Tech. Comm.*
Ishaan Shah*, **Aakash KT***, P. J. Narayanan
- EGSR 2023 **Accelerating Hair Rendering by Learning High-Order Scattered Radiance**, *CGF, Full Paper*.
Aakash KT, Adrian Jarabo, Carlos Aliaga, Matt Jen-Yuan Chiang, Olivier Maury, Christophe Hery, P. J. Narayanan, Giljoo Nam
- I3D 2022 **Bringing Linearly Transformed Cosines to Anisotropic GGX**, *Best Paper Award*.
Aakash KT, Eric Heitz, Jonathan Dupuy, P. J. Narayanan
- EGSR 2021 **Fast Analytic Soft Shadows from Area Lights**, *Full Paper*.
Aakash KT, Parikshit Sakurikar, P. J. Narayanan
- SIGGRAPH Asia 2019 **A Flexible Neural Renderer for Material Visualization**, *Technical Brief*.
Aakash KT, Parikshit Sakurikar, Saurabh Saini, P. J. Narayanan

Achievements & Activities

- 2022 **Best Paper Award, I3D 2022.**
Received NVIDIA RTX 3090 as the prize.
- 2022 **Tertiary Reviewer, ICGIP 2022.**
- 2021 **Tertiary Reviewer, Pacific Graphics 2021.**
- 2020 **KCIS Ph.D fellowship.**
Received the KCIS Ph.D fellowship for my research.
- 2019 **Microsoft Research Travel Grant.**
Received a travel grant from Microsoft to present my work at **SIGGRAPH Asia 2019**.